Nguyen Thi Van

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Professional Summary

AI Engineer with a strong foundation in Machine Learning, Natural Language Processing (Vietnamese), and Computer Vision. Experienced in building end-to-end AI solutions, including model development, deployment, and evaluation. Skilled in working with Python-based frameworks, FastAPI, and visualization tools. Known for independent research, fast learning, and a collaborative mindset.

Education

VNU University of Science, Hanoi, 10 - 2020 | 06 - 2024
Bachelor of Data Science
- GPA: 3.6/4.0

Phenikaa School of Computing, 10 - 2024 | Present
Master of Computer Science
- GPA: 3.81/4.0

Experience

Vietnam AI Intelligence Solutions, AI Engineer 08 - 2025 | Present

1. **LLM-based Education Content Automation System**
   - Built workflows for extracting, generating, and grading educational content from multimodal data.
   - Trained and optimized a YOLO model to detect and segment question regions from documents, improving system accuracy.
   - Integrated LLMs (OpenAI, Gemini) for question generation, answer evaluation, and feedback generation.
   - Applied prompt engineering techniques (few-shot, output format constraints) to improve output consistency and reduce hallucination.
2. **Vision-Language Model for Document Information Extraction**
   - Built and fine-tuned a Vision-Language Model to automate information extraction from administrative documents.
   - Designed and implemented a complete data preparation pipeline.
   - Integrated the optimized model into production using the vLLM framework for scalable inference.
3. **Symbol Detection for Document Text Extraction**
   - Developed a YOLO object detection model to automatically detect special characters in administrative documents.
   - Prepared and augmented dataset optimized for small-symbol detection.
   - Tuned and validated model performance to ensure robustness.
   - Deployed detection module as part of the text-segmentation workflow in the document processing system.
4. **Automatic Speech Recognition**
   - Conducted multimodal research on speech recognition (Vietnamese, accepted at Interspeech 2025).
   - Researched and developed speech recognition models using wav2vec2 + CTC with SpeechBrain, achieving WER 3.37 (test-clean) and 6.45 (test-other) on LibriSpeech.
5. **Evaluation of User Improvement after Product Usage (Massage Beds)**
   - Conducted controlled experiments to assess the effectiveness of massage beds.

- Analyzed and visualized collected data to derive appropriate evaluation metrics.
- Designed and implemented custom analysis functions tailored to client requirements.

BookingCare, AI Engineer — 01 - 2024 | 03 - 2024

1. **Built a medical QA chatbot using RAG architecture**
    - Built a basic RAG-based medical chatbot.
    - Processed domain-specific documents (chunking, embedding) and stored vectors in Qdrant.

Vietnam Petroleum Institute (VPI), AI Engineer Intern — 01 - 2023 | 12 - 2023

1. **Developing a Knowledge-Based QA System for the Petroleum Domain using Semantic Kernel**
    - Built a smart question-answering system for petroleum knowledge using Microsoft Semantic Kernel.
    - Implemented Python scripts to collect and extract structured text from multi-column PDFs.
    - Performed document chunking and stored data in a vector database (Qdrant) and a graph database (Neo4j).
    - Designed custom prompts and optimized LLM (GPT) responses within the Semantic Kernel framework.
    - ***Awards:*** 2nd Prize – Faculty Research Conference; Honorable Mention – University Student Research Conference
2. **Crude Oil Price Forecasting using Machine Learning and Deep Learning**
    - Role: Team leader (4 members).
    - Collected and preprocessed time-series data.
    - Engineered meaningful features from historical oil price data.
    - Applied ML and DL models (XGBoost, LSTM) to predict crude oil prices for the next 5 days.
    - Achieved RMSE 1.5 USD within the $70–82 range.
    - ***Awards:*** 3rd Prize – Faculty Research Conference; Honorable Mention – University Student Research Conference
3. **Developed a Well Management Website using Python**

Publications

ViCocktail: Automated Multi-Modal Data Collection for Vietnamese Audio-Visual Speech Recognition — Interspeech 2025

Thai Binh Nguyen, ***Thi Van Nguyen***, Quoc Truong Do, Chi Mai Luong

Projects

- Prediction of Spatial Cellular Organization from Histology Images (Top 20/ 355 Kaggle competition).
- Unraveling the Mysteries of the Female Brain: Sex Patterns in ADHD (Score = Top 2/ 1075 Kaggle competition).
- Vietnamese text summarization: Fine-tuned ViT5 model with Hugging Face; evaluated using ROUGE metrics.
- Analyzed air pollution levels in Vietnam: Collected public air quality data and built interactive dashboards using Power BI.
- Extracted vehicle speed from traffic videos: Applied OpenCV and YOLO object detection to estimate vehicle speed from real-world traffic footage.

Technical skills

- **Programming Languages:** Python (proficient), R, Java (basic), Bash.
- **AI / Machine Learning:** Built and fine-tuned models using PyTorch, TensorFlow, scikit-learn, and Hugging Face. Hands-on experience with NLP and CV projects, including both research and deployment.
- **Natural language processing:** Proficient in Vietnamese NLP: preprocessing, information extraction, and finetuning BERT/Transformer models. Familiar with RAG architecture, semantic search, and embedding-based retrieval.
- **Computer Vision:** Solid foundation in image preprocessing and CNN-based classification. Completed small-scale

projects.

- **Deployment & Visualization:** Visualized model outputs and data insights using matplotlib, seaborn, Plotly, and Power BI.
- **Other Tools:** Git, Qdrant, Neo4j, Linux.

Soft skills

- **Research & Problem Solving:** Comfortable working independently, exploring new models, reading papers, and designing applied AI solutions.
- **Teamwork & Communication:** Strong collaboration in cross-functional teams; able to communicate complex ideas clearly and take initiative.

Awards

Mathematics Talent Development Scholarship 2023 | 2024, Học kỳ I, II

Awarded to top 150 students in mathematical and data science-related majors by the Vietnam Institute for Advanced Study in Mathematics.

5x Academic Excellence Scholarship 2020 | 2024

Top 5 in GPA in Data Science, VNU University of Science.